A Modular Framework for Multi-Robot Adaptive Ocean Monitoring

with Gaussian-Process Residual Correction

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Problem & Gap

Multi-robot ocean monitoring requires combining a forecasted field with adaptive observations from a small fleet to track evolving phenomena (eddies, fronts, blooms).

Two threads of research, kept separate:

- **Classical IPP** fits a Gaussian process (GP) over the field with no physics prior *(Krause '08, Kemna '17, Jakkala '23)*.
- **Learned forecast models** (FNO, OceanNet, FourCastNet) produce skillful multi-day predictions but do not assimilate sparse mobile observations.

Gap: no general framework integrates *any* dynamical prior with multi-robot adaptive sampling, and no IPP work evaluates with the spectral metrics the learned-forecast literature now treats as standard.

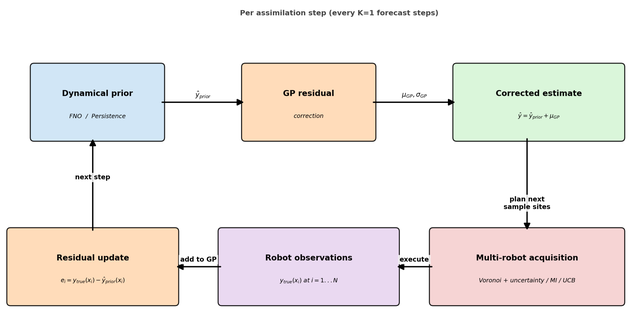
Contributions

(C1) A modular closed-loop framework. Any prior → GP residual correction → Voronoi multi-robot acquisition → observe → refit. Demonstrated with FNO, persistence, and a no-prior ablation.

(C2) A spectral evaluation methodology. 1D Hanning-windowed PSD over ocean-only transects, plus structural metrics (ACC, FSS, HF energy ratio). *We show RMSE alone is misleading in this domain.*

(C3) Empirical analysis across 4 kernels × 3 acquisitions × 4 robot counts × 5 seeds with paired statistical tests. GP loop helps any prior; learned priors win on structural metrics; no-prior methods cannot recover fine-scale structure regardless of tuning.

Closed-Loop Framework



At each step s: **(1)** prior $\hat{y}_{\text{prior}}^{(s+1)} = f(\hat{y}^{(s)})$, **(2)** GP fits residual $e(x) = y_{\text{true}}(x) - \hat{y}_{\text{prior}}(x)$, **(3)** each robot picks the next sample site within its Voronoi cell, **(4)** robots execute under glider speed budget (0.5 m/s), **(5)** corrected estimate $\hat{y} = \hat{y}_{\text{prior}} + \mu_{\text{GP}}$ feeds back. *Same code runs FNO, persistence, and no-prior.*

Method Components

Dynamical priors:

- **FNO** (OceanNet, trained on 27 yrs of ROMS SSH; PEC integrator).
- **Persistence**: $f(\hat{y}) = \hat{y}$, naive baseline.
- **No prior** (ablation): classical-IPP setup.

GP design: Matérn $\nu \in \{0.5, 1.5, 2.5\}$ or RBF. Length scale from $k(r_{\text{spacing}}) = 0.4$ where $r_{\text{spacing}} \approx 1/\sqrt{n}$.

Acquisition: uncertainty (default), GP-UCB, greedy MI. Voronoi recomputed every step.

Spectral Evaluation Methodology

Why standard PSD fails: the land mask zero-fills, creating spurious high-k content.

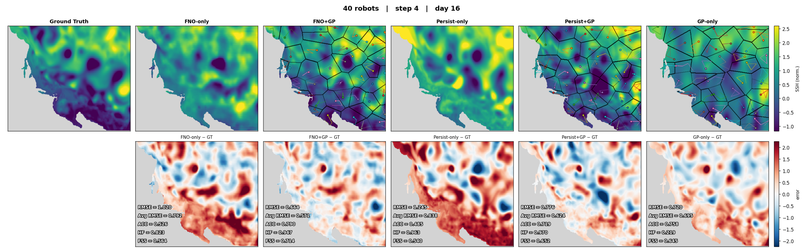
Our protocol (following Xu '22, Richman '12 for SSH analysis): for each ocean row of length $N \geq 50$ grid cells, apply Hanning window $w[n]$ and compute

$$P(k) = \frac{1}{N} \left| \mathrm{FFT}(w \cdot y)[k] \right|^2,$$

then average across all such rows. Compare $P_{\hat{y}}(k)$ vs $P_{\text{true}}(k)$ on log–log axes.

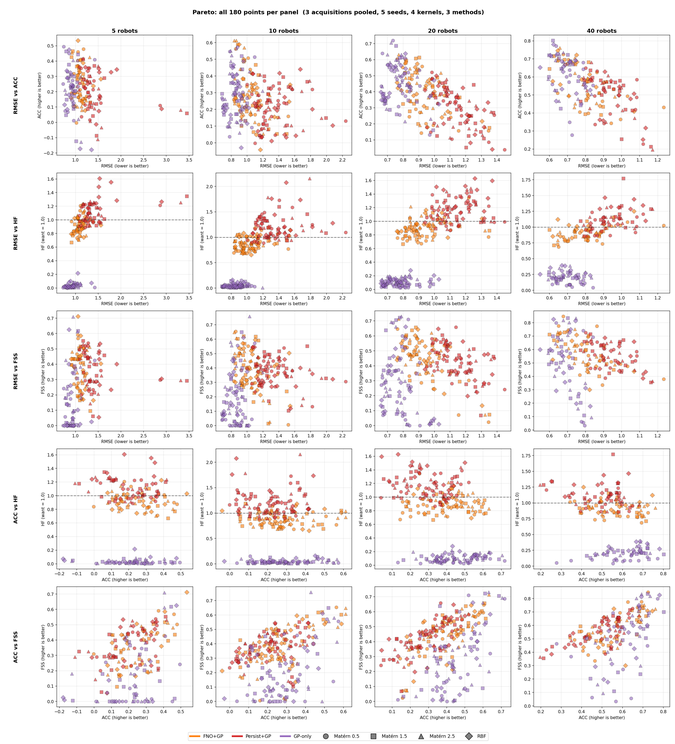
Scalar metrics: RMSE, ACC, FSS (window 5×5), and HF = ratio of high-k summed power vs ground truth (ideal = 1, smooth estimator ≈ 0).

Qualitative: GP-only is a Smooth Blob



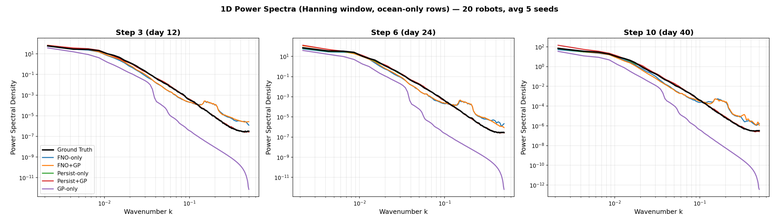
Day 16, n=40, Matérn ν=1.5, uncertainty acquisition, seed 0. GP-only resembles the climatological mean; FNO+GP and Persist+GP recover mesoscale eddies and fronts.

Hyperparameter Sensitivity (Ablation)



Each marker = one (kernel × acquisition × seed) configuration across all 4 robot counts and 5 metric pairs. Methods occupy distinct metric regions; rankings are robust to kernel and acquisition choice. The framework's qualitative outcome is not an artifact of any single hyperparameter.

Spectra: What RMSE Hides



1D Hanning-windowed PSD over ocean transects. **GP-only collapses 4–5 orders of magnitude at high** k; FNO+GP and Persist+GP track GT across the full wavenumber range. *The same property that makes GP-only a smooth blob makes it win on RMSE.*

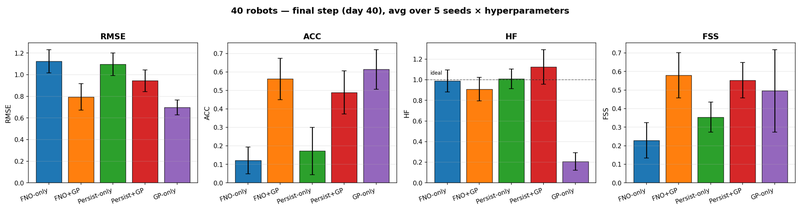
Headline Finding

GP-only "wins" RMSE *but* is a smooth blob

Method	RMSE ↓	HF (ideal = 1)
GP-only	**0.70**	0.21
FNO+GP	0.80	**0.91**
Persist+GP	0.74	1.12

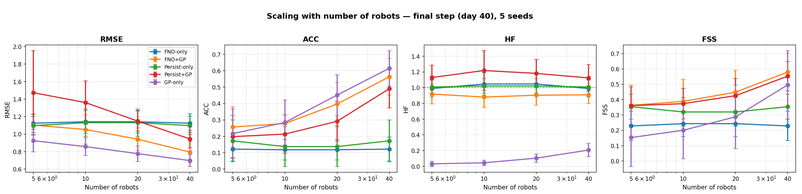
n=40, day 40, mean over 5 seeds and hyperparameters. *RMSE rewards mean-reversion; HF exposes it.*

Per-Method Comparison (n=40)



RMSE / ACC / HF / FSS, mean ± std across 5 seeds and all hyperparameters. *Read RMSE alone: GP-only wins. Read HF alone: Per+GP wins. Only the joint reading is faithful.*

Scaling with Robot Count



Framework benefit grows monotonically with observation density. Minimum useful regime ≈ 10 robots. From $n \geq 10$, FNO+GP outperforms uncorrected FNO across every metric.

Key Take-Aways

- A modular closed-loop framework integrating *any* dynamical prior with GP residual correction and multi-robot Voronoi acquisition.
- Demonstrated with FNO and persistence; the same code runs both.
- RMSE systematically rewards mean-reversion; spectral metrics expose the failure mode invisible under RMSE.
- GP loop helps any prior; learned priors yield better structural results.